

AI Application Development Guide

Chapter 1: Large Language Models

LLMs are neural networks trained on massive text corpora.

Popular models in 2026: DeepSeek-V3, Qwen3, GPT-5, Claude Opus 4.8.

Chapter 2: Prompt Engineering

System prompts define AI role, rules, and output format.

Structured JSON output is the divide between AI apps and AI chat.

Chain of Thought: let AI think step by step before answering.

Chapter 3: RAG - Retrieval Augmented Generation

RAG solves LLM knowledge cutoff and hallucination problems.

Five steps: Load -> Split -> Embed -> Store -> Retrieve.

Chunk size 300-800 recommended. Overlap 10-20% of chunk size.

Embedding converts text to vectors. Baillan text-embedding-v2 = 1536 dims.

Chapter 4: AI Agents

Agents can use tools, plan multi-step tasks, and reflect on results.

Function Calling is the standard way for LLMs to invoke external tools.

Common agent tools: search, calculator, database, API calls.